

Transformer-Based Tree Extraction from Remote Sensing Imagery: A Comprehensive Review

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Abstract

The Paris Agreement and UN SDG 15 demand scalable tree-level monitoring to quantify forest carbon stocks and ecosystem services. CNNs provide strong baselines but struggle with long-range dependencies in complex canopy structures, a limitation transformers address through self-attention, though their effectiveness remains unclear. This survey evaluates transformers on three critical requirements: superior accuracy, cross-environment generalization, and data-efficient training. Synthesizing 63 papers (2017-2025) with 550% publication growth, we analyze five architectural paradigms: pure Vision Transformers, CNN-transformer hybrids, hierarchical variants, Foundation Models, and Vision-Language Models, spanning continents and sensor modalities. Transformers yield +2.6% to +5.5% median gains over CNN baselines depending on architecture, while Foundation Models with parameter-efficient fine-tuning match full fine-tuning using 1,000 or fewer samples. Critical gaps remain: 28% average performance degradation under cross-environment transfer and 72% of studies lacking reproducibility. We contribute an architectural taxonomy, quantified performance comparisons, and practical deployment guidelines for advancing transformer-based tree extraction in geoscientific Earth observation.

Keywords: deep learning, Earth observation, remote sensing, vision transformers, tree extraction, forest monitoring

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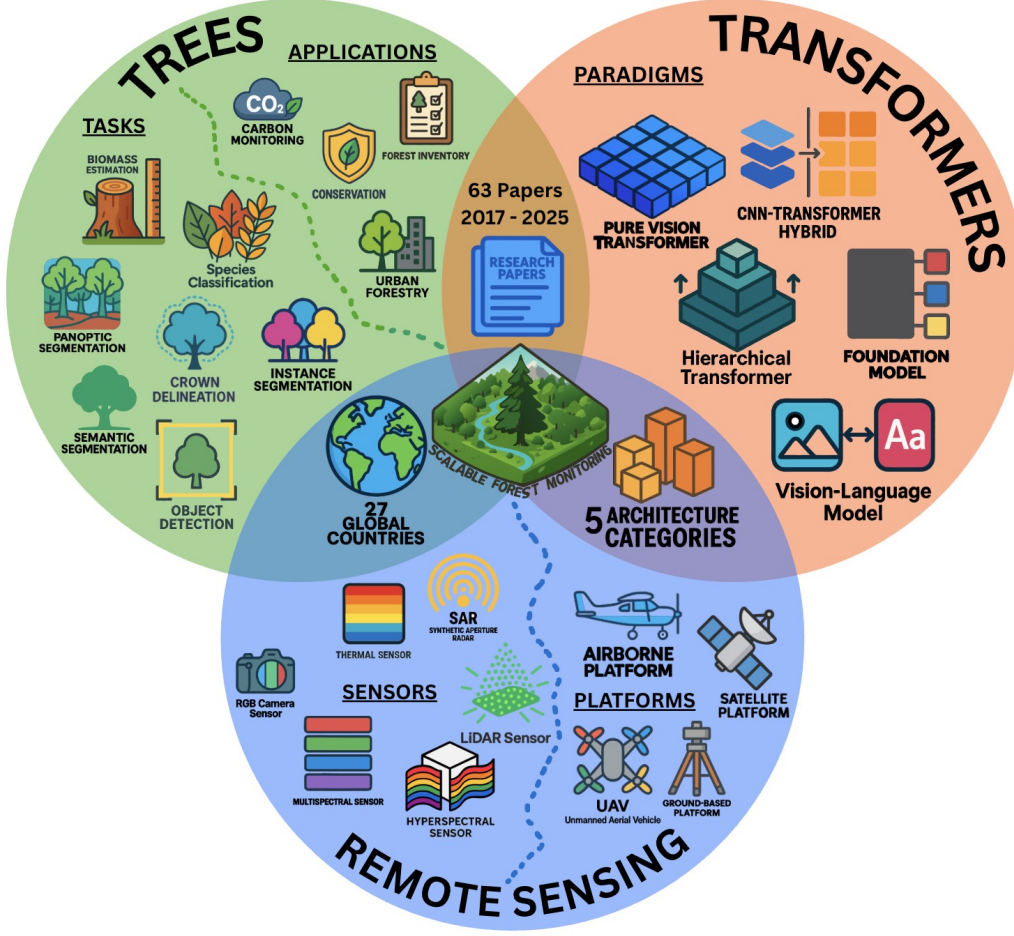


Figure 1: Overview of tree extraction survey scope.

1. Introduction

Individual tree monitoring has emerged as a critical capability for achieving global sustainability targets. The United Nations (UN) 2030 Agenda for Sustainable Development (Lee et al., 2016), particularly Sustainable Development Goal (SDG) 13 (Climate Action) and SDG 15 (Life on Land), explicitly recognizes the role of forests in climate mitigation and biodiversity conservation, driving demand for scalable, accurate tree-level monitoring systems. Forests are central to meeting climate targets under the Paris Agreement, with credible mitigation requiring precise quantification of forest carbon stocks (Grassi et al., 2017), while global tree restoration potential analyses estimate that restoring degraded lands could capture substantial atmospheric carbon (Bastin et al., 2019). National and international frameworks including Reducing Emissions from Deforestation and Forest Degradation (REDD+) require monitoring at unprecedented spatial granularity. In Europe, the European Union (EU) Nature Restoration Law mandates ecosystem recovery across degraded landscapes, creating binding obligations for member states to restore forest ecosystems (Penca and Tănăsescu, 2025). Urban greening initiatives similarly depend on individual tree inventories to model ecosystem services; tools such as i-Tree

enable municipalities to quantify carbon sequestration, air pollution removal, and stormwater management benefits at the individual tree level (Nowak, 2024). These policy imperatives have accelerated research into automated tree extraction methods that can scale across diverse landscapes while maintaining the precision needed for regulatory compliance and carbon accounting.

Traditional Remote Sensing (RS) techniques, while valuable, often struggle with the scalability and feature engineering required for precise tree-level analysis across large, heterogeneous landscapes. Convolutional Neural Networks (CNNs) marked a significant leap forward, establishing strong baselines for automated tree detection and segmentation (Zhao et al., 2023). However, CNNs' inherent focus on local features limits their ability to model the complex structures and long-range spatial dependencies characteristic of diverse forest canopies and urban environments. Vision Transformers (ViTs) have emerged as a promising alternative, employing self-attention mechanisms that directly model relationships between any two image regions regardless of spatial distance (Khan et al., 2022b). Their adoption in RS has been rapid (Aleissae et al., 2023), with particularly promising results in forestry applications including hierarchical transformers for tree detection (Gibril

Table 1: Existing review paper coverage matrix.

Review Paper	Tree Extr.	Transf.	FMs	VLMs	Note
Zheng et al. (2025)	●	◐	◐	○	Optical ITCD
Abreu-Dias et al. (2025)	●	◐	○	○	Species classif.
Lu et al. (2025)	◐	●	●	○	General RS
Tao et al. (2025)	◐	●	●	●	General RS
Wang et al. (2024a)	◐	●	◐	◐	General RS
Zhong et al. (2024)	●	◐	○	○	Species classif.
Li et al. (2024)	○	●	●	◐	Segmentation
Aleissae et al. (2023)	◐	●	◐	◐	General RS
Zhao et al. (2023)	●	○	○	○	CNN only
Chehreh et al. (2023)	●	○	○	○	Agroforestry
Liu et al. (2023)	○	●	●	●	Computer vision
Estrada et al. (2023)	●	○	○	○	Forest health
Khan et al. (2022b)	○	●	●	●	Computer vision
Diez et al. (2021)	●	○	○	○	UAV only
Velasquez-Camacho et al. (2021)	●	○	○	○	Urban trees
THIS SURVEY	●	●	●	●	Tree + Transformer

Legend: ● = Fully covered; ◐ = Partially covered; ○ = Not covered
Abbreviations: Tree Extr. = Tree Extraction; Transf. = Transformers, classif. = classification

et al., 2022) and Foundation Models (FMs) enabling tree crown segmentation with minimal labeled data (Teng et al., 2025b).

This survey addresses a critical gap in the literature. While existing reviews have addressed tree extraction methods (Zhao et al., 2023; Zheng et al., 2025) and others have surveyed transformers in RS broadly (Aleissae et al., 2023; Wang et al., 2024a), no documented synthesis comprehensively covers their intersection: transformer-based architectures, including FMs and Vision-Language Models (VLMs), for individual tree extraction across diverse environments. A rapidly growing body of research is applying transformers to tree extraction tasks, spanning instance segmentation, species classification, multimodal fusion, and FM adaptation, generating insights that require dedicated analysis. This survey comprehensively synthesizes this emerging literature, examining studies spanning multiple continents and biomes, varied sensor modalities, and a wide range of transformer architectures from pure ViTs to CNN-transformer hybrids and FM adaptations.

This survey addresses three research questions: (1) What are the performance trade-offs between transformer-based and CNN architectures for tree extraction? (2) How well do these models generalize across biomes, sensors, and resolutions? (3) What training strategies enable effective extraction with limited labeled data? Figure 1 provides a graphical overview of the survey scope, illustrating the intersection of tree extraction tasks, transformer architectures, and remote sensing modalities. To answer these questions, we contribute:

1. A data landscape analysis revealing geographic biases and reproducibility challenges across 63 papers;
2. An architecture taxonomy categorizing transformer approaches and documenting their temporal evolution;

3. An explicit performance comparison quantifying transformer advantages over CNN baselines; and

4. An analysis of FM adaptation strategies identifying when few-shot methods outperform fully-supervised specialists.

The remainder of this paper reviews related work (Section 2), analyzes datasets and reproducibility (Section 3), presents our architecture taxonomy (Section 4), addresses each research question through quantitative analysis (Section 5), identifies challenges and future directions (Section 6), and concludes with recommendations (Section 7).

2. Related Work

This survey addresses a critical gap at the intersection of transformer architectures and tree extraction applications. As shown in Table 1, while existing reviews partially address individual dimensions, no prior work comprehensively synthesizes transformer architectures, including FMs and VLMs, specifically for tree extraction. Several reviews have focused on tree extraction methods (Diez et al., 2021; Velasquez-Camacho et al., 2021; Zhao et al., 2023; Zhong et al., 2024; Zheng et al., 2025; Abreu-Dias et al., 2025), and others have surveyed transformers in RS broadly (Khan et al., 2022b; Aleissae et al., 2023; Wang et al., 2024a; Li et al., 2024), yet no prior synthesis examines transformers specifically for individual tree extraction across diverse environments. Zhao et al. (2023) systematically reviewed 35 papers (2019-2021) on CNN-based tree crown detection, establishing performance benchmarks but pre-dating widespread transformer adoption. Similarly, Zheng et al. (2025) analyzed approximately 150 papers (2000-2023) covering traditional to deep learning approaches for

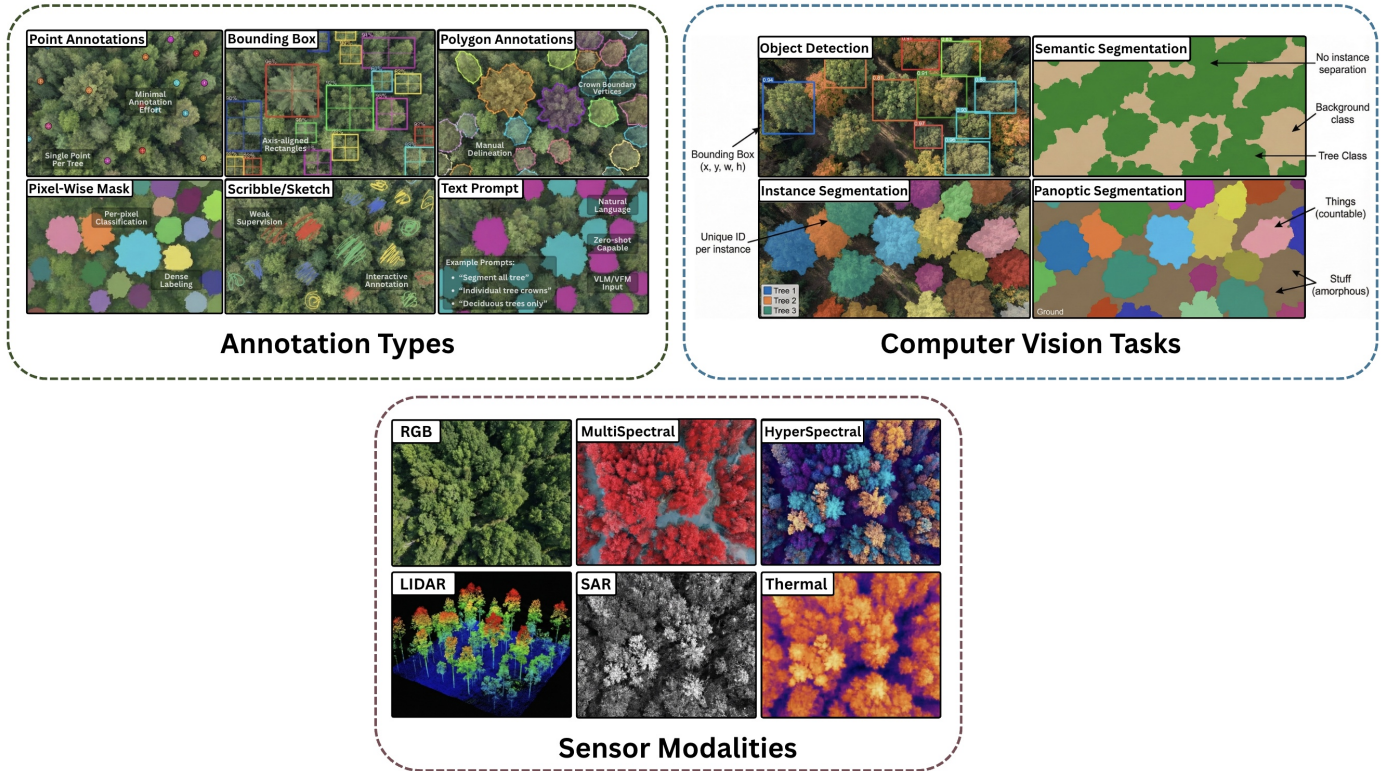


Figure 2: Overview of tree extraction research dimensions.

tree crown detection from optical imagery, with partial discussion of transformers and FMs, while task-specific reviews addressed tree species identification (Abreu-Dias et al., 2025; Zhong et al., 2024), Unmanned Aerial Vehicle (UAV)-based forestry applications (Diez et al., 2021), and urban tree characterization (Velasquez-Camacho et al., 2021). These reviews primarily document CNN achievements, with limited coverage of transformer architectures that have emerged prominently in recent years.

Conversely, transformer-focused surveys have examined these architectures across general RS and computer vision applications with limited tree-level specificity. Aleissae et al. (2023) surveyed more than 60 papers (2017-2023) on transformers for scene classification, object detection, and semantic segmentation, while Wang et al. (2024a) extended this to 237 papers (2021-2023) with systematic architectural categorization; both partially touch on vegetation-related applications but do not systematically analyze tree extraction. From the computer vision perspective, Khan et al. (2022b) and Liu et al. (2023) established theoretical foundations of vision transformers with comprehensive coverage of FMs and VLMs, while Li et al. (2024) focused on transformer-based visual segmentation including recent FM advances. While these surveys confirm transformer success in general tasks, the unique challenges of individual tree extraction, including complex canopy structures, overlapping crowns, multi-scale detection requirements, and high variability across diverse biomes, warrant focused investigation beyond general applications.

The emergence of FMs and VLMs has been documented in recent surveys. Lu et al. (2025) examined 58 FMs (2021-2024) analyzing models such as Segment Anything Model (SAM) (Kirillov et al., 2023) and Contrastive Language-Image Pre-training (CLIP) (Radford et al., 2021) for broad RS applications, with partial discussion of vegetation-related tasks, while Tao et al. (2025) reviewed approximately 90 papers (2020-2024) on VLMs for tasks including image captioning and visual question answering, also touching on environmental monitoring. Application-specific reviews have addressed agroforestry (Chehreh et al., 2023) and forest health assessment (Estrada et al., 2023). However, the systematic adaptation of these powerful models to specialized tree extraction tasks, including domain shift challenges, fine-grained species discrimination, and detection of small or sparse trees in diverse environments, represents a nascent field requiring dedicated analysis.

Our survey fills this critical gap by comprehensively analyzing 63 primary research papers (2017-2025) that apply transformer-based models (including pure ViTs, hierarchical variants, CNN-transformer hybrids, Vision Foundation Model (VFM) adaptations, and VLMs) to individual tree extraction tasks across diverse biomes, sensor modalities, and environmental contexts. Unlike general transformer surveys (Aleissae et al., 2023; Wang et al., 2024a), which partially touch on vegetation applications, we focus exclusively on tree-level analysis. Unlike tree extraction reviews (Zhao et al., 2023; Zheng et al., 2025), which primarily document CNN approaches with limited transformer discussion,

Table 2: Top public benchmarks for tree extraction with transformer models.

Dataset	Sensors	Location	Size	Res.	Used By
Open-Canopy (Fogel et al., 2025)	MS, LiDAR	France (87k km ²)	87k tiles	1.5m	Yu et al. (2025)
OAM-TCD (Veitch-Michaelis et al., 2024)	RGB	Global (>200 km ²)	280k trees	0.1m	Wilaiwongsakul et al. (2025)
Quebec Plant. (Cloutier et al., 2024)	RGB	Canada (0.44 km ²)	23k trees	0.018m	Zhang et al. (2025b); Teng et al. (2025b); Teng et al. (2025a)
Detectree2 (Ball et al., 2023)	RGB, LiDAR	Malaysia, French Guiana	3.8k trees	0.1m	Chen et al. (2025)
FOR-instanceV2 (Puliti et al., 2023)	LiDAR	5 countries	1.1k trees	500+ pts/m ²	Xiang et al. (2025)
DynamicEarthNet (Toker et al., 2022)	MS, S2	75 global sites	55k images	3m	Sadel et al. (2025)
Landcover.AI (Boguszewski et al., 2021)	RGB	Poland (216 km ²)	10.7k tiles	0.25m	Voulgaris (2025)
NeonTreeEval. (Weinstein et al., 2021)	RGB, LiDAR, HS	22 US sites	33k trees	0.1m	Xiong et al. (2025); Bountos et al. (2025); Chang et al. (2025); Chen et al. (2025); Liang (2023)
Semantic3D (Hackel et al., 2017)	Terr. LiDAR, RGB	Central Europe	4B points	0.025m	Khan et al. (2022a)
Houston/Trento (Ghamisi et al., 2015)	HS, LiDAR	US, Italy	33k px	1–2.5m	Shu et al. (2025)

Abbreviations: Res. = Resolution; MS = Multispectral; HS = Hyperspectral; S2 = Sentinel-2; Terr. = Terrestrial.

we analyze the full architectural shift to transformers and its implications for modeling complex canopy structures. Unlike FM surveys (Lu et al., 2025), we examine both custom-trained transformers and adapted FMs within a unified tree extraction framework, providing practitioners with actionable guidance for model selection and deployment.

3. Data Landscape

3.1. Benchmarks and Reproducibility

The data landscape spans diverse annotation types (bounding boxes, masks, points, polygons), computer vision tasks (object detection, semantic/instance/panoptic segmentation), and sensor modalities (Red-Green-Blue (RGB), Multispectral (MS), Light Detection and Ranging (LiDAR), Synthetic Aperture Radar (SAR), Hyperspectral (HS)). This is consolidated in Figure 2. Across 63 papers, 48.4% utilize public benchmarks (Table 2). Key datasets include NeonTreeEvaluation (Weinstein et al., 2021), the most widely adopted (Liang, 2023; Xiong et al., 2025; Chang et al., 2025; Chen et al., 2025; Bountos et al., 2025), providing 33k trees across 22 US sites with multi-modal data; FOR-instanceV2 (Puliti et al., 2023), with 1.1k tree point clouds from 5 countries at 500+ pts/m², as utilized by Xiang et al. (2025); Open-Canopy (Fogel et al., 2025), spanning 87k km² of France with 87k tiles, as utilized by Yu et al. (2025); Quebec Plantations (Cloutier et al.,

2024), with 23k trees at ultra-high 18mm Ground Sampling Distance (GSD), enabling Parameter-Efficient Fine-Tuning (PEFT) research by Zhang et al. (2025b), Teng et al. (2025b), and Teng et al. (2025a); and OAM-TCD (Veitch-Michaelis et al., 2024), offering 280k globally diverse tree annotations as leveraged by Wilaiwongsakul et al. (2025). FoMo-Bench (Bountos et al., 2025) aggregates 15 datasets for FM evaluation, while large-scale pre-training resources include MillionST (1M Sentinel-2 (S2) images) (Qin et al., 2025) and GeoLangBind-2M (2M image-text pairs) (Xiong et al., 2025).

Despite these resources, reproducibility challenges constrain progress. Across 63 papers, 64 distinct datasets are used, with 51.6% relying on private or unreported sources. Benchmark adoption remains at 25.8%, code release at 33.9% (e.g., Bountos et al., 2025; Teng et al., 2025b; Xiang et al., 2025; Chen et al., 2025), and no papers provide pre-trained transformer weights. Full reproducibility (public data and code) is achieved by only 27.4% (e.g., Soni et al., 2025; Bountos et al., 2025; Teng et al., 2025b; Qin et al., 2025). The remaining 72.6% with private data or missing code inhibit comparative evaluation and limit translation to operational systems, particularly for computationally intensive FMs.

3.2. Acquisition Platforms and Sensors

Table 3 presents a representative subset of 25 papers that collectively illustrate the diversity of platforms, sensors, and architectural approaches employed across the

Table 3: Representative subset of reviewed papers spanning all data acquisition and architectural dimensions.

Paper	Study Site	Biome	Platform	Modality	Architecture	Base Model	Task
Soni et al. (2025)	Global	Mixed	Multi	Multi	VLM	InternViT	Multi
Wilaiwongsakul et al. (2025)	Global	Mixed	Aerial	RGB	Hier. ViT	SegFormer	Seg.
Bountos et al. (2025)	Global	Forest	Multi	Multi	VFM	ViT	Multi
Gibril et al. (2025)	UAE	Semi-arid	UAV	RGB	Hier. ViT	Mask2Former	Seg.
Shu et al. (2025)	Italy, US	Mixed	Aerial	HS, LiDAR	CNN-Trans	LightViT	Cls.
Valicharla et al. (2025)	US	Temperate	UAV	RGB	Hier. ViT	Twins-SVT	Seg.
Yu et al. (2025)	France	Temperate	Satellite	MS	VFM	Swin	Multi
Perbet et al. (2024)	Canada	Boreal	Satellite	MS	Pure ViT	Transformer	Cls.
Sachdeva et al. (2024)	India	Agroforestry	Satellite	RGB	VFM	SAM	Det.
Wang et al. (2024b)	China	Mangrove	Satellite	MS	Hier. ViT	Swin	Seg.
Grondin et al. (2024)	Global	Forest	Ground	RGB	VFM	SAM	Det.
Huang et al. (2024)	US	Temperate	UAV	RGB	Pure ViT	ViT	Cls.
Amirkolaei et al. (2023)	UK, China, US	Urban	Aerial	RGB	Hier. ViT	PVT	Counting
Dersch et al. (2023)	Germany	Forest	UAV	MS, LiDAR	CNN-Trans	DETR	Seg.
Jamali et al. (2023a)	Canada	Wetland	Satellite	MS, SAR	CNN-Trans	WetMapFormer	Cls.
Du et al. (2023)	China	Urban	UAV	RGB, LiDAR	VLM	GPT-4	Multi
Jamali et al. (2023b)	Brazil	Tropical	Satellite	MS	CNN-Trans	TransU-Net++	Seg.
Sun et al. (2023)	China	Temperate	UAV	LiDAR	Pure ViT	PCT	Cls.
Gibril et al. (2023)	UAE	Agroforestry	UAV	RGB	Hier. ViT	SegFormer	Seg.
Huang et al. (2023)	China	Forest	UAV	RGB	Hier. ViT	Swin	Cls.
Fayad et al. (2023)	Ghana	Tropical	Multi	Multi	CNN-Trans	Hy-TeC	Multi
Kaselimi et al. (2022)	Brazil	Tropical	Satellite	MS	Pure ViT	ForestViT	Cls.
Gibril et al. (2022)	UAE	Semi-arid	UAV	RGB	Hier. ViT	Swin	Seg.
Khan et al. (2022a)	UK	Urban	Aerial	LiDAR	CNN-Trans	3DSVT	Multi
Zhang et al. (2022)	Brazil, China	Urban	UAV	RGB	CNN-Trans	Swin	Det.

Abbreviations: Hier. ViT = Hierarchical Vision Transformer; CNN-Trans = CNN-Transformer Hybrid; VFM = Vision Foundation Model; VLM = Vision-Language Model; Multi = Multiple platforms/modalities/tasks; Ground = Ground-based; PCT = Point Cloud Transformer; PVT = Pyramid Vision Transformer; Hy-TeC = Hybrid Transformer-ConvNet; 3DSVT = 3D Sparse Voxel Transformer; MS = Multispectral; HS = Hyperspectral; Seg. = Segmentation; Cls. = Classification; Det. = Detection.

63 reviewed studies. Sensor modality varies considerably. RGB imagery remains the most prevalent modality (42.7%), favoured for its accessibility and fine spatial detail in crown delineation and detection tasks (e.g., Zhang et al., 2022; Huang et al., 2023; Gibril et al., 2025). MS data constitutes 28.2%, offering additional spectral bands that support species discrimination, phenological monitoring, and vegetation health assessment (e.g., Kaselimi et al., 2022; Perbet et al., 2024; Wang et al., 2024b). LiDAR accounts for 16.4%, valued for its capacity to capture three-dimensional canopy structure for biomass estimation and individual tree delineation (e.g., Khan et al., 2022a; Sun et al., 2023). The remaining studies employ SAR (8.2%) for weather-independent monitoring (e.g., Jamali et al., 2023a), HS imagery (3.6%) for fine-grained spectral classification (e.g., Shu et al., 2025), and thermal sensors (0.9%) for canopy stress detection (e.g., Soni et al., 2025). Notably, over 60% of papers leverage multi-modal data fusion, exemplified by studies combining RGB with LiDAR (Du et al., 2023; Dersch et al., 2023) or MS with SAR (Jamali et al., 2023a), a trend that directly motivates transformer cross-attention mechanisms for learning complementary feature representations.

Platform selection reflects a fundamental scale-resolution trade-off that shapes downstream architectural choices. UAV platforms dominate at 52%, typically acquiring sub-decimeter RGB imagery (0.05–0.5 m GSD) suited to individual crown segmentation and detection (e.g., Huang et al., 2023; Gibril et al., 2025; Valicharla et al., 2025); this fine spatial resolution pairs naturally with hierarchical transformers (Swin, SegFormer) whose multi-scale feature extraction captures both local texture and broader canopy context. Satellite platforms constitute 31%, primarily deploying MS and SAR sensors for landscape-scale monitoring (e.g., Kaselimi et al., 2022; Jamali et al., 2023b; Perbet et al., 2024), where coarser spatial granularity motivates pure ViTs or CNN-transformer hybrids capable of processing larger spatial contexts. Aerial platforms (15%) bridge this gap at 0.25–1.0 m GSD, serving regional forest inventories with balanced spatial coverage and resolution (e.g., Khan et al., 2022a; Amirkolaei et al., 2023). Ground-based systems, though represented by only a single study (Grondin et al., 2024), provide millimetre-precision 3D structural characterisation that enables foundation model adaptation for fine-grained tree segmentation. This platform-sensor distribution underscores how multi-modal fusion tasks drive specialised cross-modal attention architectures that learn complementary representations, specifically texture from RGB and structure from LiDAR, through attention-based alignment, a key transformer advantage over CNN feature concatenation, further examined in Section 4.

3.3. Geographic Distribution

Geographic analysis of the 63 reviewed studies reveals pronounced spatial concentration. While 28.6% draw on datasets spanning multiple countries, over 80% of single-country studies are situated in the Northern Hemisphere.

The United States accounts for the largest share at 12.7%, encompassing crown delineation, species mapping, and height estimation across diverse forest types. China follows at 11.1%, then the UAE at 9.5% with a distinctive focus on date palm mapping in arid landscapes. Brazil (9.5%) anchors tropical forest research (Kaselimi et al., 2022; Jamali et al., 2023b), Canada (7.9%) contributes work on boreal disturbance detection and crown segmentation (Jamali et al., 2023a; Perbet et al., 2024), and Germany (6.3%) advances detection and segmentation in temperate European forests (Dersch et al., 2023). Individual studies cover France, India, Bolivia, Italy, the UK, and Ghana, while synthetic or unspecified study areas account for 6.3%.

Biome representation exhibits a similar imbalance. General forest environments dominate at 44.4%, followed by temperate forests at 19.0%, urban environments (9.5%), agroforestry (9.5%), and tropical forests (9.5%). Critically underrepresented biomes include boreal forests (1.6%) (Perbet et al., 2024), despite comprising roughly 29% of global forest area; mangroves (1.6%) (Wang et al., 2024b); wetlands (1.6%) (Jamali et al., 2023a); and semi-arid drylands (1.6%) (Gibril et al., 2025). This concentration carries tangible methodological consequences: models validated primarily on temperate and general forest canopies may struggle with the multi-layered structural complexity of tropical forests, the spectral ambiguity of mangrove environments, the pronounced seasonal dynamics of boreal systems, or the sparse tree distributions characteristic of semi-arid landscapes. Cross-biome validation remains limited, with only 24% of studies evaluating performance across ecologically distinct settings.

4. Transformer Architectures

4.1. Temporal Trends

Figure 3 traces the temporal evolution of transformer adoption in tree extraction, revealing over 550% growth from 2022 to 2025. Although our comprehensive review scope spans 2017–2025 to capture the full transformer era following the original Transformer architecture (Vaswani et al., 2017), the first applications to tree extraction appeared in 2022, reflecting the time required for vision transformers to mature and for the RS community to adapt these architectures to forestry applications. Early adoption in 2022 established the viability of transformer approaches through pioneering work on Swin Transformer for date palm detection (Gibril et al., 2022), Detection Transformer (DETR) for single-tree detection (Dersch et al., 2022), 3D LiDAR transformers for urban vegetation (Khan et al., 2022a), and individual tree detection from UAV imagery (Zhang et al., 2022).

The field expanded rapidly in 2023, establishing CNN-Transformer hybrids as the dominant paradigm. Notable contributions include TransUNet variants for deforestation mapping (Jamali et al., 2023b,a), canopy height estimation (Fayad et al., 2023), species classification (Wang et al., 2023;

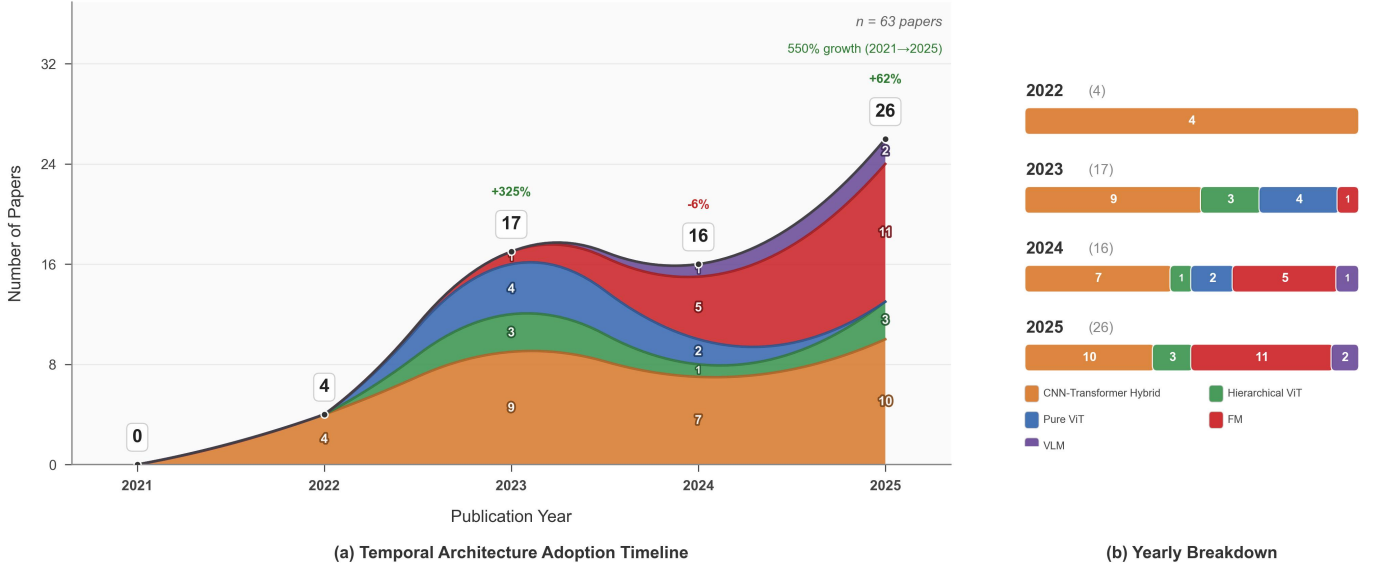


Figure 3: Temporal adoption of different transformer architectures.

Huang et al., 2023; Sun et al., 2023), optical-SAR fusion (Ferrari et al., 2023; Mullissa et al., 2023), DETR-based detection refinement (Dersch et al., 2023), cross-sensor benchmarking (Gominski et al., 2023), and the first VLM application with Tree-GPT (Du et al., 2023). The year 2024 marked a qualitative shift toward FMs, with SAM adaptations (Grondin et al., 2024; Liu et al., 2024a; Speckenwirth et al., 2024; Muszynski et al., 2024) and distribution-shift benchmarking (Sachdeva et al., 2024) emerging alongside continued hybrid refinement through e-TransUNet (Jamali et al., 2024) and multi-modal fusion networks (Lu et al., 2024; Zhu et al., 2024; Wang et al., 2024b). This FM wave accelerated dramatically in 2025, with SAM-based crown segmentation (Teng et al., 2025b,a; Chen et al., 2025; Wasil et al., 2025), parameter-efficient fine-tuning frameworks (Zhang et al., 2025b), multi-modal FMs (Bountos et al., 2025; Xiong et al., 2025; Yu et al., 2025), and interactive VLMs for forest dialogue (Soni et al., 2025). Hybrids remained substantial with innovations in multimodal fusion (Zhang et al., 2025a; Zhou et al., 2025; Shu et al., 2025; Chang et al., 2025) and 3D point cloud processing (Xiang et al., 2025), while hierarchical transformers saw continued use for date palm detection (Shanableh et al., 2025; Begum et al., 2025) and invasive species mapping (Valicharla et al., 2025; Joshi and Witharana, 2025).

Three trends define the field. First, CNN-Transformer hybrids dominate (27.0%), balancing established CNN backbones with transformer mechanisms. Second, FM adaptations are surging. Absent in 2022, they became the fastest-growing category by 2024–2025, fueled by accessible pre-trained models and efficient tuning methods like Low-Rank Adaptation (LoRA). Third, VLM applications are emerging (4.8%) but face computational and spatial reasoning hurdles. This shift toward FMs suggests a move from custom architectures to adapting large-scale models,

offering better generalization and reduced annotation needs (see RQ3).

4.2. Architectural Taxonomy

Analysis of 63 papers reveals five primary architectural categories that trace the evolution from early ViT adaptations to contemporary FM approaches: Pure ViT, CNN-Transformer Hybrids, Hierarchical ViTs, FMs, and VLMs. An additional 28.6% employ custom or incompletely reported backbones and are grouped as Unspecified/Custom in Figure 5. The distribution reflects both the maturity of different approaches and their suitability for tree-level analysis in RS contexts. Conceptual processing pipelines for each category are illustrated in Figure 4.

4.2.1. Pure Vision Transformers

Pure ViT architectures apply standard ViT (Dosovitskiy et al., 2020) and sequence-to-sequence formulations such as Segmentation Transformer (SETR) (Zheng et al., 2021) to tree extraction (7.9%), treating inputs as sequences of fixed-size patches or tokens processed by global self-attention (Kaseliimi et al., 2022; Mullissa et al., 2023; Pierdicca et al., 2023; Huang et al., 2024; Sun et al., 2023). In this literature, pure ViT models are primarily used for classification-oriented tasks, including deforestation monitoring via multilabel classification (Kaseliimi et al., 2022) and tree species classification from UAV canopy imagery (Huang et al., 2024; Pierdicca et al., 2023) and LiDAR-derived inputs (Sun et al., 2023). Sequence-based transformers are also applied to seasonal disturbance detection from multi-sensor time series (Mullissa et al., 2023). Overall adoption remains limited relative to hybrid and hierarchical designs, reflecting the data-hungriness and compute cost of training fully attention-based models in RS settings.

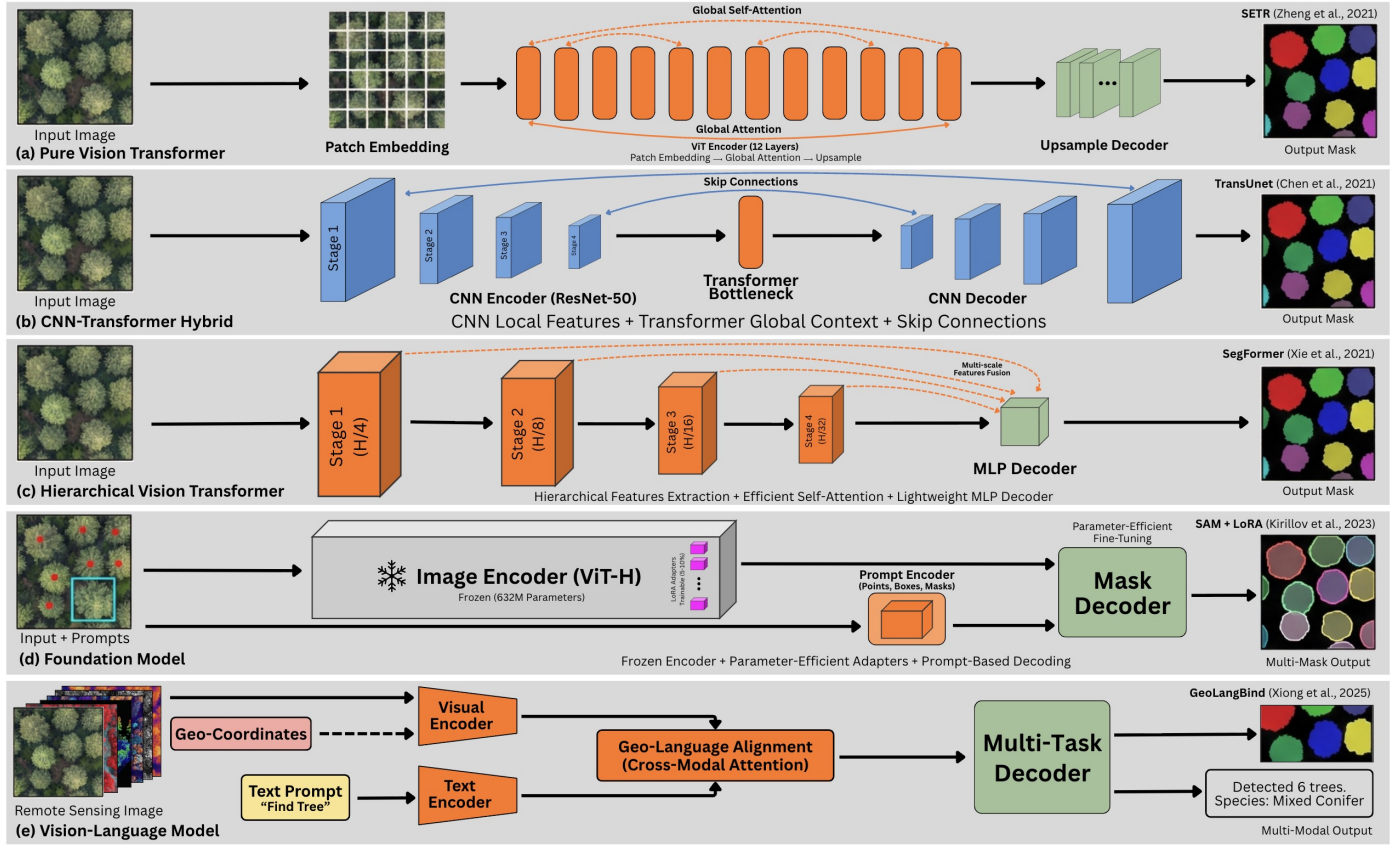


Figure 4: Transformer architectural categorization pipelines.

4.2.2. CNN-Transformer Hybrids

CNN-Transformer hybrids comprise 27.0% of the literature, pairing CNN encoders for efficient local feature extraction with transformer decoders or attention modules for global context modeling (e.g., Jamali et al., 2024; Dersch et al., 2023; Zhang et al., 2025a; Xiang et al., 2025). This hybrid design exploits complementary strengths: CNNs provide translation equivariance and hierarchical feature pyramids, while transformers contribute long-range dependency modeling that helps separate overlapping crowns and capture larger-scale spatial structure. ResNet-based feature pyramids remain the most common backbone pattern, often feeding transformer modules for delineation or dense prediction (Gibril et al., 2024; Wang et al., 2025). The TransUNet family (Chen et al., 2021) adapts U-Net skip connections with transformer bottlenecks for RS segmentation (Jamali et al., 2024). Detection transformers (DETR) (Carion et al., 2020) replace hand-crafted pipelines with learned object queries for single-tree detection (Dersch et al., 2022, 2023), while Mask2Former (Cheng et al., 2022) supports instance-aware crown delineation in high-resolution imagery (Wang et al., 2025) and LiDAR-derived canopy height models (Yang et al., 2023). Domain-specific hybrids address multi-modal sensing and 3D structure: MTCNet fuses RGB and LiDAR cues for crown detection (Zhang et al., 2025a), and ForestFormer3D extends hybrid modeling to point-cloud segmentation (Xiang et al., 2025).

4.2.3. Hierarchical Vision Transformers

Hierarchical ViTs, particularly Swin Transformer (Liu et al., 2021) and SegFormer (Xie et al., 2021) (11.1%), employ hierarchical feature extraction without CNN backbones (e.g., Gibril et al., 2023; Joshi and Witharana, 2025; Valicharla et al., 2025; Begum et al., 2025). Unlike standard ViT's global self-attention with quadratic complexity, hierarchical variants compute attention within local windows, reducing complexity while maintaining long-range dependency modeling through shifted window mechanisms. Other hierarchical backbones that appear in the broader vision literature include PVT (Wang et al., 2021), DaViT (Ding et al., 2022), UniFormer (Li et al., 2023), and Twins-SVT (Chu et al., 2021). SegFormer (Xie et al., 2021) combines hierarchical encoders with lightweight Multilayer Perceptron (MLP) decoders for applications including unhealthy tree crown detection (Joshi and Witharana, 2025) and invasive species mapping (Valicharla et al., 2025), while Twins-SVT (Chu et al., 2021) supports efficient inference for fine-grained vegetation monitoring (Valicharla et al., 2025). As in Figure 5, papers that embed hierarchical transformers inside CNN-based pipelines (e.g., Mask R-CNN + Swin) are treated as hybrids to keep categories mutually exclusive.

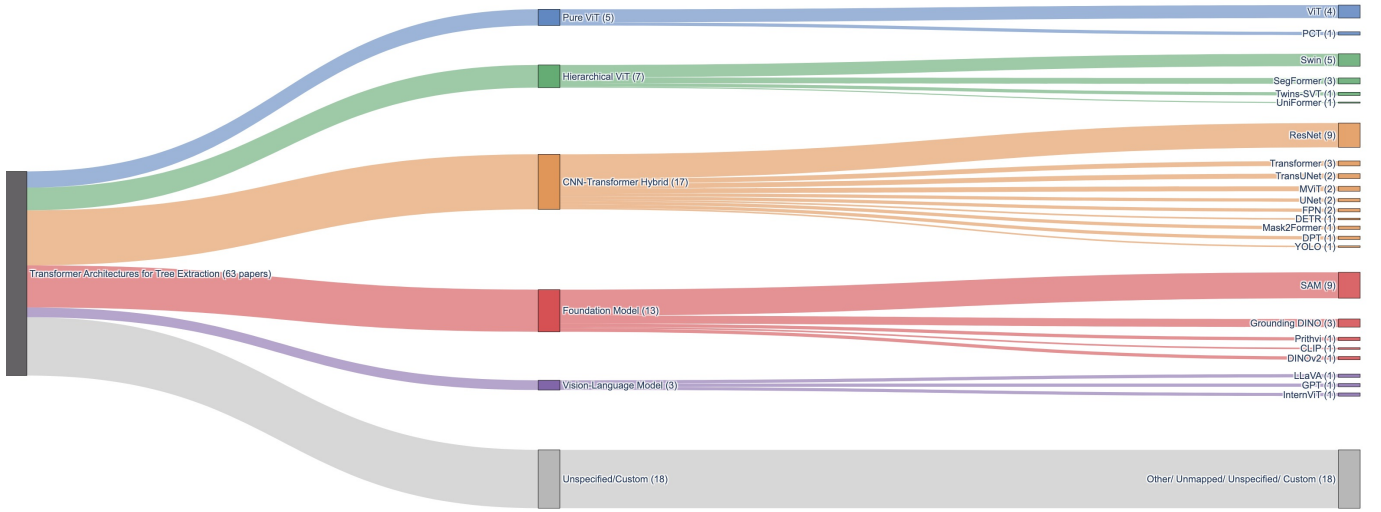


Figure 5: Transformer architecture taxonomy for tree extraction.

4.2.4. Foundation Models

FM adaptations account for 20.6% and represent the fastest-growing direction, leveraging large pre-trained models such as SAM (Kirillov et al., 2023) and Grounding DINO (Liu et al., 2024b) for tree segmentation and detection with reduced task-specific training (e.g., Teng et al., 2025b; Chen et al., 2025; Wasil et al., 2025; Sachdeva et al., 2024). SAM-based studies explore a spectrum of adaptation strategies: zero-shot SAM2 evaluation without domain training yields low precision but moderate recall (0.41-0.65) depending on the benchmark (Chen et al., 2025), while parameter-efficient fine-tuning can reach near full fine-tuning performance with a tiny fraction of trainable parameters (Zhang et al., 2025b) and can match strong adapter baselines with substantially fewer parameters using LoRA (Wasil et al., 2025). Fully supervised adaptation and prompt-learning variants further improve crown delineation but increase computational cost (Teng et al., 2025b,a; Grondin et al., 2024; Liu et al., 2024a). Grounding DINO supports open-vocabulary or prompt-guided tree detection, enabling species-aware detection and downstream classification under weak supervision (Sachdeva et al., 2024; Zhang et al., 2025b). Overall, FMs directly target a central bottleneck in tree extraction: the high cost of pixel-level annotation, by leveraging representations learned from large-scale natural-image or RS-specific pretraining corpora.

4.2.5. Vision-Language Models

VLMs represent an emerging frontier (4.8%), enabling interactive and multi-task capabilities through natural language interfaces (Soni et al., 2025; Du et al., 2023; Xue et al., 2024). EarthDial (Soni et al., 2025) turns multi-sensory earth observations into interactive dialogues for

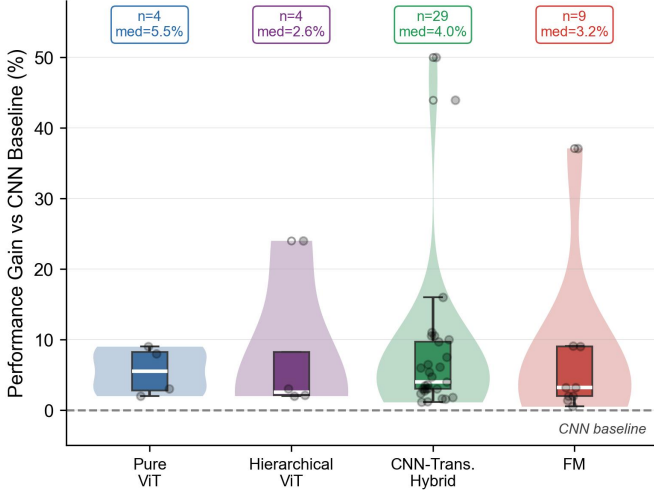
querying forest conditions and species composition, while Tree-GPT (Du et al., 2023) implements a modular Large Language Model (LLM) expert system combining specialized visual encoders with LLM reasoning. REO-VLM (Xue et al., 2024) addresses regression challenges through language-guided visual grounding for quantitative attribute estimation (e.g., height, biomass, canopy cover). While VLMs promise more accessible forest monitoring through natural language interfaces, computational requirements and limitations in fine-grained spatial reasoning currently constrain operational deployment.

5. Discussion

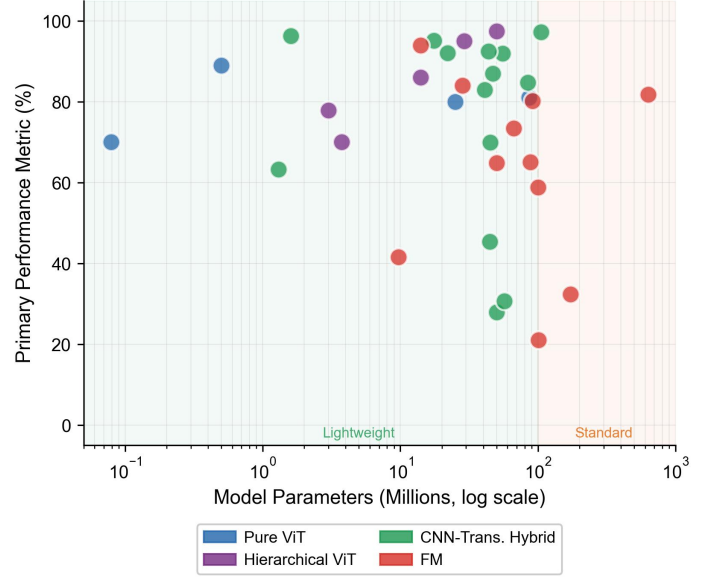
5.1. RQ1: Performance Trade-offs

Transformer architectures consistently outperform CNN baselines across tree extraction tasks (e.g., Gibril et al., 2022; Dersch et al., 2023; Jamali et al., 2024; Teng et al., 2025b; Bountos et al., 2025). Across 61 transformer-based studies (excluding two non-transformer papers; Wang et al., 2023; Jia et al., 2025), five architectural categories exhibit distinct performance profiles, with median gains computed from studies reporting CNN baseline comparisons on percentage-based metrics (Intersection over Union (IoU), Average Precision (AP), F1, accuracy) as visualized in Figure 6(a).

CNN-transformer hybrids constitute the largest category (30 papers, 29 with CNN baselines) and deliver a robust median gain of +4.0% across segmentation and detection, leveraging CNN encoders for local features and transformer decoders for global context (e.g., Dersch et al., 2023; Jamali et al., 2024; Zhang et al., 2025a; Xiang et al.,



(a) Performance Gain Distribution by Architecture



(b) Performance vs. Computational Cost

Figure 6: Performance-efficiency trade-offs across transformer architectures.

2025; Wang et al., 2025). This complementary design proves particularly effective for crown delineation, where boundary precision and long-range spatial relationships both matter. Pure ViTs (6 papers, 4 with baselines) achieve the highest median improvement (+5.5%) but with limited sample size, concentrating in species classification, health assessment, invasive species mapping, and spatiotemporal modeling (e.g., Huang et al., 2024; Joshi and Witharana, 2025; Kasehimi et al., 2022; Pierdicca et al., 2023), though their higher parameter requirements (up to 86M) and data hunger limit adoption for smaller datasets. Hierarchical ViTs (6 papers, 4 with baselines) show modest gains (+2.6%) across classification and segmentation.

FM adaptations (15 papers, 9 with baselines) exhibit a median improvement of +3.2% but fundamentally trade raw performance for label efficiency, primarily targeting segmentation and detection (e.g., Teng et al., 2025b; Chen et al., 2025; Wasil et al., 2025; Bountos et al., 2025). This category includes papers adapting pre-existing FMs (SAM, Grounding-DINO, Prithvi) as well as self-built FMs such as TiMo (Qin et al., 2025). Zero-shot SAM performance varies widely across domains, from 65% recall on the Detectree2 benchmark (Chen et al., 2025) to 33% F1 on multispectral UAV data where a conventional Faster R-CNN substantially outperforms (Speckenwirth et al., 2024), underscoring that pre-trained representations do not universally transfer to tree extraction. PEFT approaches narrow this gap: adapters and learned prompts achieve competitive segmentation and detection results while training only 0.28–2.2% of model parameters (Zhang et al., 2025b; Liu et al., 2024a; Wasil et al., 2025), making them the most annotation-efficient strategy currently available. VLMs (4 papers; Du et al., 2023; Xue et al., 2024; Soni et al., 2025;

Xiong et al., 2025) report no CNN baselines since they address fundamentally different interactive tasks such as visual question answering, dialogue, and language-guided regression, though computational requirements and limited spatial reasoning currently constrain deployment.

Figure 6(b) plots model parameters against primary performance metrics, delineating efficiency frontiers: lightweight architectures (<100M parameters) achieve competitive performance (70–97%) while larger FMs (>100M) yield marginal accuracy gains at substantially higher computational cost. Notable outliers below 50% reflect metric heterogeneity rather than model failure: studies reporting COCO-style mAP averaged across IoU thresholds 0.5–0.95 yield values approximately half their AP_{50} counterparts (e.g., 45.4% mAP vs. 93.1% AP_{50} ; Zhang et al., 2025a), boundary F1 at strict tolerance thresholds penalizes even minor spatial offsets (Gominski et al., 2023), and zero-shot SAM without domain adaptation achieves only 21% IoU on multispectral imagery (Speckenwirth et al., 2024). Four mechanisms underpin the consistent transformer advantage: *long-range dependency modeling* for capturing spatial relationships between distant image regions, critical for distinguishing overlapping crowns (Gibril et al., 2022; Dersch et al., 2023; Zhou et al., 2025); *multi-scale feature extraction* through hierarchical architectures (Gibril et al., 2023; Al-Ruzouq et al., 2024; Huang et al., 2024); *adaptive receptive fields* via dynamic attention weights (Wang et al., 2025; Valicharla et al., 2025); and *end-to-end learning* eliminating hand-crafted post-processing (Dersch et al., 2022, 2023).

These advantages, however, come at computational cost: transformers incur approximately $2.3\times$ median parameter counts relative to CNN baselines, with model sizes span-

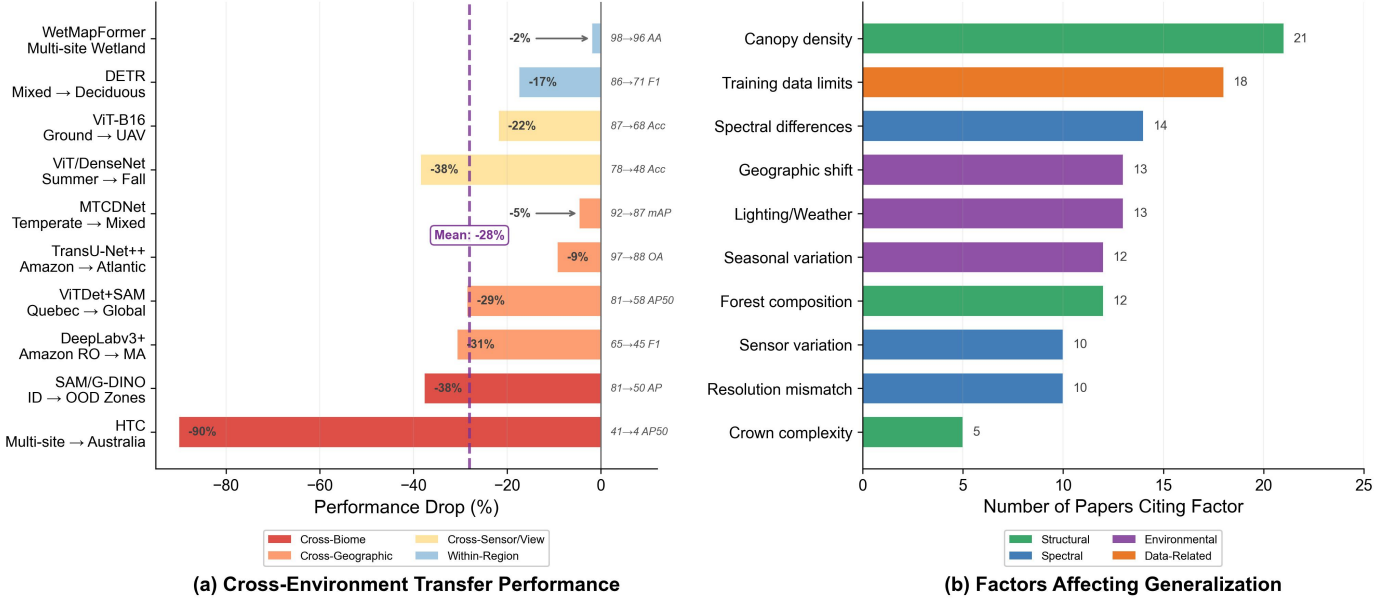


Figure 7: Cross-environment generalization.

ning three orders of magnitude from lightweight hybrids such as WetMapFormer (1.6M; Jamali et al., 2023a) to LLaVA-based VLMs (13B; Xue et al., 2024). This wide span reveals an efficiency paradox for FMs: despite requiring the largest parameter budgets (66M–632M), pre-trained models enable effective extraction with minimal labeled data, potentially offsetting their computational overhead through reduced annotation costs (Teng et al., 2025b; Grondin et al., 2024). In practice, this creates two deployment regimes: lightweight hierarchical transformers (3.75–116M parameters) suit resource-constrained UAV platforms where inference latency is critical, while larger FMs are better matched to cloud-based offline analysis where accuracy justifies the computational investment.

5.2. RQ2: Cross-Environment Generalization

Cross-environment generalization emerges as a critical challenge for operational deployment (e.g., Gominski et al., 2023; Grondin et al., 2024; Sachdeva et al., 2024; Chen et al., 2025; Bountos et al., 2025). Figure 7(a) synthesizes transfer performance across 10 representative experiments, revealing an average 28% performance drop whose severity varies systematically with ecological distance: cross-biome transfers incur a mean 64% degradation, cross-geographic transfers within similar biomes a mean 18% drop, cross-sensor and viewpoint shifts approximately 30%, and within-region transfers only 10%.

Cross-biome transfers exhibit the most severe failures. The HTC model (Wang et al., 2025), trained on 9 globally distributed sites, achieves 41% AP50 on average but drops to 4% on unseen Australian savanna, while SAM with Grounding-DINO shows 20–40% degradation on out-of-distribution agro-climatic zones (Sachdeva et al., 2024).

Notably, CNNs can outperform pure ViTs in such cross-domain scenarios: DeepLabv3+ generalized better than UNETR in 7 of 9 cross-domain combinations across Brazilian tropical biomes (Vega et al., 2024). Cross-geographic transfers within ecologically similar biomes yield more favorable outcomes: TransU-Net++ degrades only 9% from Amazon to Atlantic Forest (Jamali et al., 2023b, 2024), MTCDNet maintains performance within 5% between Chinese temperate and mixed forests (Zhang et al., 2025a), and ViTDet+SAM reveals a striking asymmetry whereby training on Quebec alone and testing globally yields 29% relative degradation (Figure 7a), whereas training on globally diverse FLICKRTREE data reduces Quebec-specific degradation to only 4 percentage points (Grondin et al., 2024). This last result demonstrates that diverse training exposure can enable environment-invariant feature learning.

Cross-sensor and viewpoint shifts introduce a distinct category of distribution mismatch (e.g., Pierdicca et al., 2023; Mullissa et al., 2023; Huang et al., 2024; Xiong et al., 2025). Ground-level classifiers trained on iNaturalist suffer 22% degradation when applied to UAV aerial imagery (Pierdicca et al., 2023), and seasonal variation proves particularly challenging, with models trained on summer imagery achieving only 48% accuracy on fall data (Mullissa et al., 2023; Huang et al., 2024). SAM2 offers a partial counterpoint, demonstrating notable viewpoint invariance by generalizing from California temperate to Malaysian tropical forests without fine-tuning (Chen et al., 2025), though with moderate recall (41–65%). Within-region transfers incur the mildest degradation: DETR-based detection drops only 17% from mixed to deciduous stands within the same German site (Dersch et al., 2022, 2023), and WetMapFormer maintains 96% accuracy across New Brunswick

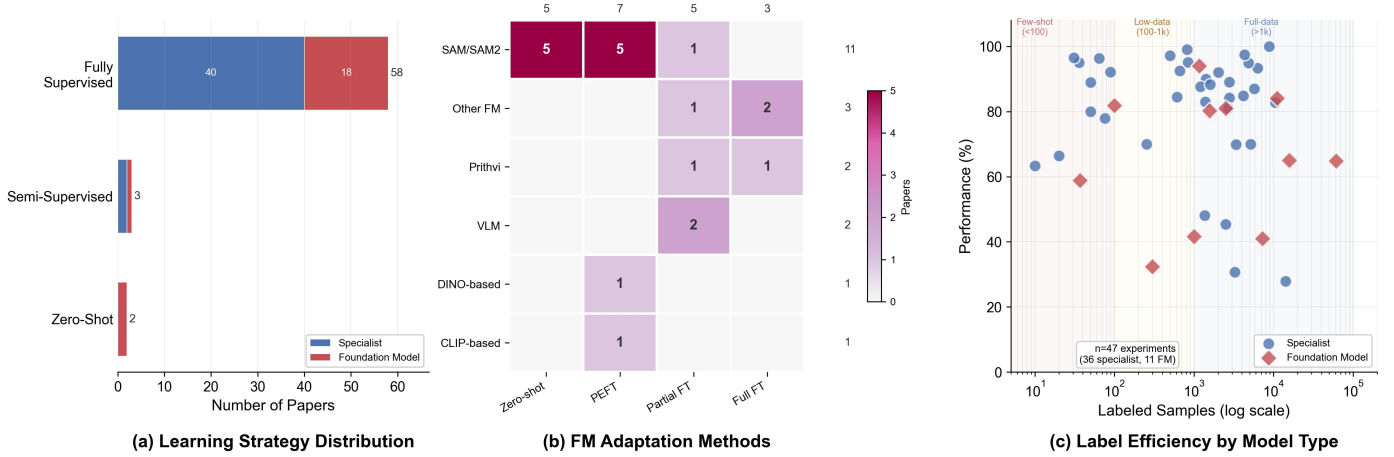


Figure 8: Training strategy and data efficiency analysis.

wetland sites (Jamali et al., 2023a). Collectively, these patterns indicate that architectural sophistication alone cannot overcome fundamental domain mismatches, but FMs pre-trained on diverse imagery offer a promising path toward robust generalization (Bountos et al., 2025; Teng et al., 2025b,a; Chen et al., 2025; Xiong et al., 2025; Qin et al., 2025).

Figure 7(b) categorizes the factors driving these generalization failures by citation frequency. Structural factors dominate: canopy density emerges as the primary challenge, since models trained on open woodland fail on dense tropical canopy (e.g., Gibril et al., 2022; Gominski et al., 2023; Dersch et al., 2023; Xiang et al., 2025), while forest composition and crown complexity also degrade performance substantially (Wang et al., 2025; Zhou et al., 2025; Shu et al., 2025; Voulgaris, 2025). Spectral factors, including signature differences and resolution mismatch (Amirkolaei et al., 2023; Fayad et al., 2023; Sachdeva et al., 2024; Ton-That et al., 2024), and sensor variation (e.g., Ferrari et al., 2023; Gui et al., 2025; Chen et al., 2025), disproportionately affect models relying on specific band combinations. Environmental factors such as geographic shift, illumination conditions, and seasonal variation (e.g., Mullissa et al., 2023; Huang et al., 2024; Al-Ruzouq et al., 2024) introduce systematic appearance differences. Among all factor categories, training data diversity limitations (e.g., Grondin et al., 2024; Bountos et al., 2025; Wasil et al., 2025) are the most directly addressable through careful dataset curation and FM pre-training on globally diverse imagery.

5.3. RQ3: Data-Efficient Training

Training strategy selection significantly impacts both model performance and deployment feasibility, particularly under limited annotation budgets (e.g., Teng et al., 2025b; Wasil et al., 2025; Chen et al., 2025; Bountos et al., 2025). Figure 8(a) reveals that fully supervised learning dominates (92%), with FMs concentrated in emerging alternatives: semi-supervised (5%) and zero-shot (3%). Importantly, this

panel classifies papers by their *downstream* training strategy; papers employing self-supervised pre-training (e.g., masked autoencoding) followed by supervised fine-tuning are classified as “Fully Supervised,” and a paper that uses SAM in zero-shot mode for pseudo-label generation while employing supervised training for other components is likewise classified as “Fully Supervised.” Figure 8(b) examines FM-level adaptation methods across 20 papers employing foundation models, showing that SAM/SAM2 dominates (11 papers) with zero-shot (5) and PEFT (5) being equally common, followed by partial fine-tuning (1). Other FM families include DINO-based (1), CLIP-based (1), Prithvi (2), VLMs (2), and custom FMs such as FoMo-Net and TiMo (3). The apparent discrepancy between panels (2 zero-shot papers in Figure 8(a) versus 5 SAM zero-shot instances in Figure 8(b)) arises because three papers use zero-shot SAM as a component within supervised or semi-supervised pipelines (Grondin et al., 2024; Wang et al., 2025; Speckenwirth et al., 2024). Figure 8(c) assesses data efficiency empirically, plotting labeled sample count against performance for 47 experiments reporting both specific sample counts and percentage-based metrics (filtering out non-comparable metrics such as MAE/RMSE); the resulting high variance (28–100% at similar sample counts) reflects heterogeneous evaluation settings and task complexity rather than intrinsic model deficiency. The low-performance cluster (<50%) is largely attributable to strict metric choices: COCO-style mAP averaged across IoU thresholds 0.5–0.95 produces values roughly half the corresponding AP₅₀ (e.g., 45% mAP vs. 93% AP₅₀; Zhang et al., 2025a; Zhu et al., 2024), and multi-class mIoU is depressed by challenging minority classes such as coarse woody debris (Wasil et al., 2025; Voulgaris, 2025).

FM adaptation represents the fastest-growing strategy category, dominated by SAM-based approaches. Table 4 catalogues 12 adaptation experiments across five base model families; because each targets different tasks, datasets, and evaluation metrics, cross-row comparison should emphasize qualitative patterns over absolute scores.

Table 4: FM Adaptation Strategies for Tree Extraction.

Base Model	Adaptation	Train. (%)	Samples	Perf.	Task	Reference
SAM2 (Hiera-L)	Zero-shot	0	0	0.65 Recall	Seg., Det.	Chen et al. (2025)
SAM	Zero-shot	0	0	33% F1	Seg.	Speckenwirth et al. (2024)
SAM (ViT-H)	Zero-shot (prompt eng.)	0	~100	60.6 AP_{50}^{seg}	Seg., Det.	Grondin et al. (2024)
SAM (ViT-H)	PEFT (Adapter)	1.5	1,000	41.6% mIoU	Seg.	Wasil et al. (2025)
SAM (ViT-H)	PEFT (DSMPrompter)	NR	15,742	64.84% wmAP	Seg., Cls.	Teng et al. (2025a)
SAM (ViT-H)	PEFT (BalSAM)	NR	61,327	64.84% wmAP	Seg.	Teng et al. (2025b)
SAM (ViT-H), CLIP	PEFT (adapter)	~2.2	1,160	0.94 mIoU	Seg., Cls., Bio.	Liu et al. (2024a)
G-DINO (Swin-T)	PEFT (ASCS)	0.28	300	60.6 AP_{50}	Det., Cls.	Zhang et al. (2025b)
G-DINO, SAM	Head-only FT	NR	~2,500	82.1 AP_{50}	Det.	Sachdeva et al. (2024)
Prithvi-100M	Full FT	100	37 AOIs	0.963 Dice	Change det.	Sadel et al. (2025)
DOFA-CLIP	PEFT (adapter)	NR	2.05M	59.0% mIoU	Cls., Seg.	Xiong et al. (2025)
FoMo-Net (ViT-B)	Full FT	100	NR	95.24% F1	Cls., Seg., Det.	Bountos et al. (2025)

Abbreviations: Train. (%) = trainable parameters as percentage of total; Perf. = Performance; NR = Not Reported; ~ = estimated; PEFT = Parameter-Efficient Fine-Tuning; ASCS = Adaptive Sparse Channel Selection; FT = Fine-Tuning; wmAP = weighted mean Average Precision; AOIs = Areas of Interest; Seg. = Segmentation; Det. = Detection; Cls. = Classification; Ret. = Retrieval; Bio. = Biomass Estimation.

Zero-shot application of FMs provides the lowest entry barrier but reveals a fundamental limitation: performance depends critically on alignment between pretraining distribution and target domain. SAM2 achieves 65% recall on the Detectree2 benchmark for tropical and temperate crowns (Chen et al., 2025), while vanilla SAM yields only 33% F1 on multispectral UAV imagery, where a Faster R-CNN trained on the same data reaches 83.5% (Speckenwirth et al., 2024). This gap confirms that FMs pre-trained on natural images lack the spectral and structural priors needed for reliable tree extraction across sensor modalities, and that CNNs remain competitive baselines when sufficient in-domain training data is available. Strategic prompt engineering can narrow this gap without modifying model weights: by training a lightweight burn-in detector on ~100 labeled images and feeding its predictions as structured prompts to frozen SAM, Grondin et al. (2024) achieve 60.6 AP_{50}^{seg} for tree trunk instance segmentation across geographically diverse forests.

PEFT methods occupy the most active region of the adaptation design space. A recurring finding is that adapter architecture and domain-specific auxiliary signals matter more than sample count. With only 300 labeled images and 0.28% of parameters trainable, Adaptive Sparse Channel Selection on Grounding-DINO reaches 60.6 AP_{50} for joint tree crown detection and species classification (Zhang et al., 2025b). Incorporating geospatial context proves especially valuable: DSMPrompter and BalSAM both achieve 64.84% wmAP for crown instance segmentation by integrating elevation data into SAM’s prompting pathway (Teng et al., 2025a,b), and a multi-FM pipeline combining SAM with CLIP reaches 0.94 mIoU for urban vegetation segmentation while training ~2.2% of parameters (Liu et al., 2024a).

These results suggest that the key bottleneck for PEFT in tree extraction is not parameter efficiency itself, but the integration of domain knowledge, such as canopy elevation, spectral indices, and structural context, that natural-image pre-training does not capture.

These patterns yield a practical decision framework. With scarce labels (<100 samples), zero-shot FMs with prompt engineering provide a viable starting point (Grondin et al., 2024). With moderate annotation budgets (100–1,000 samples), PEFT methods outperform training from scratch, particularly when adapter design incorporates auxiliary geospatial inputs (Wasil et al., 2025; Zhang et al., 2025b). With abundant labels, fully supervised specialists can achieve higher peak in-distribution performance, though head-only fine-tuning of Grounding-DINO outperforms full fine-tuning under distribution shift (Sachdeva et al., 2024), suggesting that minimal adaptation better preserves generalizable representations. Semi-supervised approaches occupy an intermediate niche: TreeFormer achieves approximately 90% of fully supervised counting performance using only 30% labeled data (Amirkolaee et al., 2023).

Pre-training corpus composition further modulates downstream effectiveness. Although ImageNet pre-training remains dominant, RS-specific pre-training consistently outperforms it for forestry applications. FoMo-Net, pre-trained via masked autoencoding on multi-modal RS archives, achieves 95.24% F1 on forest-type classification (Bountos et al., 2025), and Prithvi-100M, pre-trained on Harmonized Landsat Sentinel (HLS) data, reaches 0.963 Dice for forest change detection from Sentinel-2 time series (Sadel et al., 2025). By contrast, DOFA-CLIP’s vision-language pre-training on 2.05M image-text pairs yields 59.0% mIoU on

the NeonTreeEvaluation segmentation benchmark (Xiong et al., 2025), highlighting a persistent gap when transferring general-purpose vision-language representations to fine-grained spatial tasks that demand precise boundary delineation. Self-supervised pre-training via Masked Autoencoders (MAEs) on unlabeled RS imagery shows particular promise for reducing annotation requirements (Muszynski et al., 2024; Bountos et al., 2025). VLMs extend tree extraction capabilities through natural language interfaces (Du et al., 2023; Soni et al., 2025) and language-guided regression for quantitative attribute estimation (Xue et al., 2024), though their computational requirements currently constrain operational deployment.

6. Challenges and Future Directions

6.1. Current Challenges

Three interconnected challenges constrain the transition from experimental demonstrations to operational forest monitoring systems.

The *capability-efficiency tension* presents the most immediate barrier to deployment. Transformers impose approximately $2.3\times$ higher parameter counts than CNN baselines, with FMs such as SAM ViT-H requiring 632M parameters (Teng et al., 2025b; Chen et al., 2025). Lightweight alternatives, notably SegFormer-B0 (3.75M parameters) and WetMapFormer (1.6M), demonstrate that competitive accuracy is achievable at substantially reduced computational cost (Joshi and Witharana, 2025; Jamali et al., 2023a), yet systematic benchmarking of accuracy-efficiency trade-offs across deployment scenarios remains absent. Practitioners thus navigate an under-characterized design space, choosing between cloud-based inference (with attendant latency and connectivity constraints) and edge deployment (with reduced model capacity).

Cross-environment generalization constitutes the most consequential scientific limitation. Our analysis documents 28% average performance degradation outside training distributions, with cross-biome transfers producing severe failures (Sachdeva et al., 2024; Grondin et al., 2024; Wang et al., 2025). The root causes are systemic: 85% of studies concentrate in the Northern Hemisphere, boreal forests (comprising 29% of global forest area) receive only 1.6% of research attention, and merely 22% of studies validate across ecologically distinct environments with documented performance outcomes. These biases imply that published accuracy figures likely overestimate operational capability for the majority of global forests.

The *reproducibility deficit* compounds both preceding challenges. Across the 63 reviewed papers, only 27.4% achieve full reproducibility (public data and code), while 51.6% rely on private datasets, benchmark adoption stands at 25.8%, code availability at 33.9%, and no study releases pre-trained transformer weights (Bountos et al., 2025; Voulgaris, 2025). This opacity inhibits systematic comparison and constrains the translation of research advances into

deployed systems. Moreover, although FMs have unlocked few-shot and zero-shot capabilities, 92% of papers still require fully supervised training (Amirkolaee et al., 2023; Teng et al., 2025b). PEFT achieves approximately 99% of full fine-tuning performance with 1,000 or fewer labeled samples (Wasil et al., 2025; Zhang et al., 2025b), yet principled guidance on sample selection strategies and active learning protocols remains absent.

6.2. Future Research Directions

For practitioners navigating immediate deployment decisions, we offer architecture-selection guidance grounded in our analysis: with limited labels (<100 samples), FMs with PEFT achieve near full fine-tuning accuracy at reduced annotation cost; with moderate data (100–1,000 samples), semi-supervised approaches offer favorable performance-cost trade-offs; and with abundant labels ($>1,000$ samples), specialist hybrid architectures maximize accuracy. Deployment constraints further shape this decision: lightweight hierarchical transformers suit edge and UAV platforms, while larger FMs are better suited to cloud-based processing pipelines.

For the research community, four priorities emerge from the gaps identified in this survey. First, *globally representative benchmarks* spanning at least eight biomes with standardized multi-modal acquisition protocols are essential to redress the geographic concentration documented in Section 3.3. Second, *systematic domain adaptation research*, encompassing unsupervised methods, continual learning, and meta-learning, is needed to close the 28% generalization gap without requiring per-site annotation. Third, *model compression and distillation* techniques must be advanced to enable transformer deployment on UAV and portable platforms where inference budgets are tightly constrained. Fourth, *multi-modal fusion* architectures exploiting cross-modal attention mechanisms and self-supervised pre-training on unlabeled RS imagery offer a path toward more robust and data-efficient representations.

Sustained progress ultimately depends on stronger reproducibility norms: mandatory validation across ecologically distinct environments, systematic release of code and pre-trained weights, and transparent reporting of computational requirements. A centralized model repository for tree extraction, analogous to existing efforts in medical imaging and natural language processing, would reduce duplicated effort and accelerate the translation of research prototypes into operational monitoring systems.

7. Conclusion

This survey comprehensively analyzes transformer-based approaches for tree extraction, reviewing 63 papers (2017–2025). The field has experienced 550% growth from 2022 to 2025, evolving from pure ViTs through hybrids to FM adaptations. Key findings: transformers consistently outperform CNN baselines ($+4.0\%$ median gain for hybrids)

with architectural choices mattering more than scale; models exhibit 28% average performance drops across domain shifts, with FMs pre-trained on diverse data demonstrating better robustness; and FMs with PEFT offer the best path for limited-label scenarios while specialist architectures maximize accuracy with abundant data.

This survey contributes a data landscape analysis revealing geographic bias (85% Northern Hemisphere, 27.4% fully reproducible), an architecture taxonomy documenting the shift toward FMs, quantified performance comparisons, and practical deployment recommendations. Priority directions include multi-biome benchmarks, domain adaptation research, efficient edge architectures, and reproducibility standards. Realizing transformer potential for global forest monitoring requires addressing limitations in generalization, efficiency, and open science to accelerate progress toward scalable, geoscience-ready tree extraction systems supporting sustainability goals.

CRedit authorship contribution statement

Attavit Wilaiwongsakul: Conceptualization, Methodology, Data curation, Formal analysis, Investigation, Visualization, Writing (original draft). **Bin Liang:** Supervision, Project administration, Writing (review). **Bryan Zheng:** Supervision, Conceptualization, Writing (review). **Fang Chen:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No new data were generated or analysed in this study. All reviewed papers are cited in the reference list. Supplementary materials, including the full paper catalogue and analysis scripts, are publicly available at <https://github.com/attavit14203638/transformer-tree-survey>.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used Claude (Anthropic) in order to improve language, statistical verification, and figure generation. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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